MA1521 Cheatsheet

Ch 0: Real Numbers & Functions

Properties

- **Triangle Inequality:** $|x+y| \le |x| + |y|$
- **Absolute Value:**
  - $|-x| = |x|$
  - $|xy| = |x||y|$
  - $\sqrt{x^2} = |x|$
  - $|x| < r \iff -r < x < r$ (for $r > 0$)

Ch 1: Limits and Continuity

Theorems & Laws

- **Limit Laws:** For sums, differences, products, and quotients (denom $\neq 0$), the limit of the operation is the operation of the limits.
- **Squeeze (Sandwich) Theorem:** If $g(x) \le f(x) \le h(x)$ near c and $\lim_{x\to c} g(x) = \lim_{x\to c} h(x) = L$, then $\lim_{x\to c} f(x) = L$.
- **Intermediate Value Theorem (IVT):** If f is continuous on $[a,b]$ and k is between $f(a)$ and $f(b)$, there exists $c \in [a,b]$ such that $f(c) = k$.

Special Trig Limits

- **Small Angle Approximation:**

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \quad \text{and} \quad \lim_{x\to 0} \frac{1-\cos x}{x} = 0$$

- **Note:** $\lim_{n\to\infty} \frac{\sin n}{n} = 0$ (by Squeeze Thm).
- *Application:* $\lim_{n\to\infty} n \sin\left(\frac{k}{n}\right) = k$.

Continuity

f is continuous at c if:

$$\lim_{x\to c} f(x) = f(c)$$

Ch 2: Derivatives

Definition

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Table of Derivatives

$(x^n)' = nx^{n-1}$	$(\cot x)' = -\csc^2 x$
$(e^x)' = e^x$	$(\sec x)' = \sec x \tan x$
$(\ln x)' = 1/x$	$(\csc x)' = -\csc x \cot x$
$(\sin x)' = \cos x$	$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}}$
$(\cos x)' = -\sin x$	$(\tan^{-1} x)' = \frac{1}{1+x^2}$
$(\tan x)' = \sec^2 x$	$(\sec^{-1} x)' = \frac{1}{\lvert x\rvert\sqrt{x^2-1}}$

Parametric Differentiation

For $x = f(t), y = g(t)$:

$$\frac{dy}{dx} = \frac{g'(t)}{f'(t)} \quad \text{and} \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(y')}{\frac{dx}{dt}}$$

Ch 3: Applications of Differentiation

Lines & Approximations

- **Tangent Line:** $y - y_0 = m(x - x_0)$ where $m = f'(x_0)$.
- **Normal Line:** $y - y_0 = -\frac{1}{m}(x - x_0)$.
- **Linear Approximation (Linearization):**

$$f(x) \approx L(x) = f(a) + f'(a)(x-a)$$

- **Newton's Method (Root Finding):**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Theorems

- **Rolle's Theorem:** If f is continuous on $[a,b]$, differentiable on (a,b), and $f(a) = f(b)$, there is a $c \in (a,b)$ where $f'(c) = 0$.
- **Mean Value Theorem (MVT):** Under the same conditions, there is a $c \in (a,b)$ where:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

- **L'Hôpital's Rule:** For forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$:

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}$$

Ch 4: Integrals

Riemann Sum Definition

The definite integral $\int_a^b f(x)\,dx$ is defined as:

$$\lim_{n\to\infty} \left\{ \sum_{k=1}^{n} \left(\frac{b-a}{n}\right) f\left(a + k\left(\frac{b-a}{n}\right)\right) \right\}$$

Fundamental Theorem of Calculus

- **Part 1:** $\int_a^b f(x)\,dx = F(b) - F(a)$ (F is antiderivative of f).
- **Part 2:** $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.

Common & Trig Integrals

- $\int \ln x\,dx = x\ln x - x + C$
- $\int \tan x\,dx = \ln|\sec x| + C$
- $\int \cot x\,dx = \ln|\sin x| + C$
- $\int \sec x\,dx = \ln|\sec x + \tan x| + C$
- $\int \csc x\,dx = \ln|\csc x - \cot x| + C$

Techniques

- **Integration by Parts:**

$$\int u\,dv = uv - \int v\,du$$

- **Substitution:**

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

Tangent Half-Angle ($z = \tan\left(\frac{\theta}{2}\right)$)

Used for rational functions of $\sin\theta, \cos\theta$:

- $\sin\theta = \frac{2z}{1+z^2}, \quad \cos\theta = \frac{1-z^2}{1+z^2}$
- $d\theta = \frac{2}{1+z^2}\,dz$
- **Limits:** $0 \to 0, \quad \frac{\pi}{2} \to 1, \quad \pi \to \infty$

Ch 5: Applications of Integration

Area Between Curves

- **Vertical Strips:** $A = \int_a^b |f(x) - g(x)|\,dx$
- **Horizontal Strips:** $A = \int_c^d |f(y) - g(y)|\,dy$

Volume of Revolution

- **Disk Method (about x-axis):**

$$V = \pi \int_a^b [f(x)]^2\,dx$$

- **Washer Method (about x-axis):**

$$V = \pi \int_a^b \left([f(x)]^2 - [g(x)]^2\right)\,dx$$

- **Shell Method (about y-axis):**

$$V = 2\pi \int_a^b x|f(x)|\,dx$$

Surface Area of Revolution

- **About the x-axis:** For $y = f(x)$ from a to b:

$$S = 2\pi \int_a^b f(x)\sqrt{1 + [f'(x)]^2}\,dx$$

- **About the y-axis:** For $x = g(y)$ from c to d:

$$S = 2\pi \int_c^d g(y)\sqrt{1 + [g'(y)]^2}\,dy$$

- **For parametric equations (about x-axis):**

$$S = 2\pi \int_c^d y(t)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$

Arc Length

- For $y = f(x)$ from a to b (and vice versa for $x = g(y)$):

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$$

- For parametric equations:

$$L = \int_c^d \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$$

Ch 6: Infinite/Convergent Series

Sum of Geometric Series

$$S_n = \sum_{i=1}^{n} a_i \cdot r^{i-1} = a_1\left(\frac{1-r^n}{1-r}\right)$$

- Diverges if $|r| \ge 1$
- Converges to $\frac{a}{1-r}$ if $|r| < 1$

Series

p-Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

- Converges if $p > 1$
- Diverges if $p \le 1$

Harmonic Series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

- **Never Converges**
- Specific case of **p-series** when $p = 1$

Test for Convergence

Integral Test

- For a series $\sum a_n$ where $a_n = f(n)$ and $f(x)$ are continuous, positive, and decreasing for $x \ge 1$:
- Converges **if and only if:**

$$\int_1^{\infty} f(x)\,dx \textbf{ converges}$$

n-th Term Test (Divergence Test)

- If $\sum a_n$ converges, then $\lim_{n\to\infty} a_n = 0$
- Diverges if not, but converse (if q, then p) is not necessarily true (e.g. harmonic series)

Comparison Test

- Suppose $0 \le a_n \le b_n$
  - If $\sum b_n$ converges, then $\sum a_n$ converges
  - If $\sum a_n$ diverges, then $\sum b_n$ diverges

Ratio Test

$$L = \lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right| \quad \text{then} \quad R = \frac{1}{L}$$

- If $L < 1$: **Absolutely Convergent**
- If $L > 1$: **Divergent**
- If $L = 1$: **Inconclusive**

Root Test

$$L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$$

- Conclusions are identical to Ratio Test

Alternating Series Test

$$\sum (-1)^{n-1} b_n \text{ with } b_n > 0$$

- If b_n is decreasing ($b_{n+1} \leq b_n$) and $\lim_{n\to\infty} b_n = 0$, then the series converges.

Power Series

$$\sum_{n=0}^{\infty} c_n \cdot (x-a)^n, \text{ centered at } a$$

- Radius of Convergence
  - The series converges abs for $|x-a| < R$ and diverges for $|x-a| > R$
  - $R = \frac{1}{L} \implies L$ is the limit from Ratio or Root test applied to coeffs
- **Take note:** Within radius of convergence, can be differentiated and integrated term-by-term

Taylor and Maclaurin Series

Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

- Representation of a function $f(x)$ as a power series centred at $x = a$

Maclaurin Series

- Taylor Series centred at $a = 0$

Important Maclaurin Series

- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$
- $(1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \dots$
- $\frac{1}{1+x} = \sum_{n=0}^{\infty}(-1)^n x^n = 1 - x + x^2 - x^3 + \dots$ (converges for $|x| < 1$)
- $ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ (converges for $-1 < x \leq 1$)
- $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$
- $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$
- $\tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ (converges for $|x| \leq 1$)

Ch 7: Vectors and Geometry of Space

3D Coordinates & Vectors

- **Distance:**

$$|P_1P_2| = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$$

- **Equation of Sphere:**

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

- **Length:** $\|\mathbf{a}\| = \sqrt{a_1{}^2 + a_2{}^2 + a_3{}^2}$
- **Unit Vector:** $\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}$

The Dot Product

- $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 = \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta$
- $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$

Projections

- **Scalar Projection (comp):**

$$\text{comp}_{\mathbf{a}}\mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|}$$

- **Vector Projection (proj):**

$$\text{proj}_{\mathbf{a}}\mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2}\right)\mathbf{a}$$

- **Point to Plane Distance:**

$$D = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2+b^2+c^2}}$$

The Cross Product

- **Definition:**

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

- **Magnitude:** $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\| \sin\theta$
- $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.
- **Point Q to Line PR Dist:** $d = \frac{\|\vec{PQ} \times \vec{PR}\|}{\|\vec{PR}\|}$

Lines & Planes

- **Line (Vector):** $\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$
- **Line (Parametric):** $x = x_0 + at$, $y = y_0 + bt$, $z = z_0 + ct$
- **Plane (Vector):** $\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$
- **Plane (Scalar):** $A(x-x_0) + B(y-y_0) + C(z-z_0) = 0$

Ch 8: Functions of Several Variables

Arc Length Formula

- Let C be: $r(t) = \langle f(t), g(t), h(t) \rangle$, $a \leq t \leq b$
- f', g' and h' are continuous

$$s = \int_a^b \sqrt{f'(t)^2 + g'(t)^2 + h'(t)^2}\, dt$$

Surface Area of a 2D Surface in 3D

- For $z = f(x,y)$ over region D:

$$S = \iint_D \sqrt{1 + (f_x)^2 + (f_y)^2}\, dA$$

Tangent Plane & Linear Approx

- **Tangent Plane:** $z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$
- **Linear Approximation:** $f(x,y) \approx f(a,b) + f_x(a,b)dx + f_y(a,b)dy$

Second Partials Test

Let $D = f_{xx}(a,b)f_{yy}(a,b) - [f_{xy}(a,b)]^2$:

- $D > 0, f_{xx}(a,b) > 0 \implies$ Local Minimum
- $D > 0, f_{xx}(a,b) < 0 \implies$ Local Maximum
- $D < 0 \implies$ Saddle Point
- $D = 0 \implies$ Test is Inconclusive

Chain Rule

- Given: $z = f(x,y)$, where $x = g(s,t)$ and $y = h(s,t)$ (all are differentiable by s and t)

$$\frac{\partial z}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s}$$

for each $i = 1, \dots, m$

Change of Variables (Jacobian)

For a one-to-one transformation $x = x(u,v)$ and $y = y(u,v)$:

$$\iint_R f(x,y)\, dA = \iint_S f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv$$

where the Jacobian J is given by:

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} f_{xu} & f_{xv} \\ f_{yu} & f_{yv} \end{vmatrix}$$

and the area elements are related by:

$$dA = |J|\, du\, dv, \qquad \frac{\partial(u,v)}{\partial(x,y)} = \frac{1}{J}$$

For 3 variables $x = x(u,v,w)$, $y = y(u,v,w)$, and $z = z(u,v,w)$:

$$\iiint_R f(x,y,z)\, dV = \iiint_S f(x(u,v,w), y(u,v,w), z(u,v,w)) \times \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\, dv\, dw$$

Common Case

Polar Coordinates

Using the transformation $x = r\cos\theta$ and $y = r\sin\theta$, the Jacobian is $|J| = r$. Therefore:

$$dA = r\, dr\, d\theta$$

Cylindrical Coordinates

Using $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, the Jacobian is

$$\left|\frac{\partial(x,y,z)}{\partial(r,\theta,z)}\right| = r.$$

Hence,

$$dV = r\, dr\, d\theta\, dz$$

Spherical Coordinates

Using

$$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi,$$

where ρ is length from origin, ϕ is angle from positive z-axis, and θ is angle from positive x-axis in xy-plane.
with $\rho \geq 0$, $0 \leq \phi \leq \pi$, $0 \leq \theta < 2\pi$, the Jacobian is

$$\left|\frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)}\right| = \rho^2 \sin\phi.$$

Hence,

$$dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$$

Ch 9: Ordinary Differential Equations

Separable Equations

- **Form:** $y' = g(x)h(y)$
- **Solution:** $\int \frac{1}{h(y)}\, dy = \int g(x)\, dx$

First-Order Linear ODEs

- **Form:** $y' + P(x)y = Q(x)$
- **Factor:** $I(x) = \exp\left(\int P(x)\, dx\right)$
- **Solution:** $y = \frac{1}{I(x)} \int I(x)Q(x)\, dx$

Bernoulli Equations

- **Form:** $y' + P(x)y = Q(x)y^n$ $(n \neq 0, 1)$
- **Substitution:** $v = y^{1-n}$
- **Linearized:** $v' + (1-n)P(x)v = (1-n)Q(x)$

Trigonometric Identities

Pythagorean

- $\sin^2 x + \cos^2 x = 1$
- $\sec^2 x - 1 = \tan^2 x$

Double Angle

- $\sin 2x = 2\sin x\cos x$
- $\cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$

Power-Reduction (Half-Angle)

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

$$\sin^2 x = \frac{1-\cos 2x}{2}$$

Product-to-Sum

$$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

$$\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$$

Special Inverse & Hyperbolic Integrals

- $\int \frac{1}{\sqrt{a^2-x^2}}\, dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{a^2+x^2}\, dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2-a^2}}\, dx = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$
- $\int \frac{1}{\sqrt{x^2+a^2}}\, dx = \ln|x + \sqrt{x^2+a^2}| + C$
- $\int \frac{1}{\sqrt{x^2-a^2}}\, dx = \ln|x + \sqrt{x^2-a^2}| + C$

Trigonometric Substitution

Used for integrals containing radicals of quadratic forms:

- **Form** $\sqrt{a^2-x^2}$: Let $x = a\sin\theta$, $dx = a\cos\theta\, d\theta$
  (Relies on $1 - \sin^2\theta = \cos^2\theta$)
- **Form** $\sqrt{a^2+x^2}$: Let $x = a\tan\theta$, $dx = a\sec^2\theta\, d\theta$
  (Relies on $1 + \tan^2\theta = \sec^2\theta$)
- **Form** $\sqrt{x^2-a^2}$: Let $x = a\sec\theta$, $dx = a\sec\theta\tan\theta\, d\theta$
  (Relies on $\sec^2\theta - 1 = \tan^2\theta$)